

Appendix A

Complete Rolling-Horizon UC Constraints

The rolling-horizon UC model minimizes the total operation cost. The model includes the following standard constraints.

Additional symbols are defined below. $P_{\text{wind},t}$, $P_{\text{pv},t}$, and $P_{\text{load},t}$ denote the forecast wind power, photovoltaic power, and load power. $P_g^{\max}$ and $P_g^{\min}$ denote the maximum and minimum generation limits of unit g . $R_t^{\min}$ denotes the minimum spinning reserve requirement. T_g^{up} and T_g^{dn} denote the minimum up and down times of unit g . R_g^{up} and R_g^{dn} denote the ramp up and ramp down rates of unit g . $F_{l,t}$ denotes the active power flow on line l . B_l and x_l denote the line susceptance and reactance. $\theta_{i,t}$ denotes the voltage phase angle at bus i . $F_l^{\max}$ denotes the thermal capacity of line l .

The power balance constraint matches total supply and served load:

$$\begin{aligned} \sum_g P_{g,t} + P_{\text{wind},t} - P_{\text{wind},t}^{\text{curt}} + P_{\text{pv},t} - P_{\text{pv},t}^{\text{curt}} \\ = P_{\text{load},t} - P_{\text{load},t}^{\text{shed}}, \quad \forall t \end{aligned} \quad (1)$$

The spinning reserve constraint preserves generation headroom:

$$\sum_g (P_g^{\max} X_{g,t} - P_{g,t}) \geq R_t^{\min}, \quad \forall t \quad (2)$$

The minimum up time and down time constraints use standard binary logic:

$$\sum_{\tau=t}^{t+T_g^{\text{up}}-1} X_{g,\tau} \geq T_g^{\text{up}}(X_{g,t} - X_{g,t-1}), \quad \forall g, \forall t \quad (3)$$

$$\sum_{\tau=t}^{t+T_g^{\text{dn}}-1} (1 - X_{g,\tau}) \geq T_g^{\text{dn}}(X_{g,t-1} - X_{g,t}), \quad \forall g, \forall t \quad (4)$$

The ramp rate constraints limit generation changes between consecutive hours:

$$\begin{aligned} P_{g,t} - P_{g,t-1} \leq R_g^{\text{up}} X_{g,t-1} \\ + P_g^{\max}(X_{g,t} - X_{g,t-1}), \quad \forall g, \forall t \end{aligned} \quad (5)$$

$$\begin{aligned} P_{g,t-1} - P_{g,t} \leq R_g^{\text{dn}} X_{g,t} \\ + P_g^{\max}(X_{g,t-1} - X_{g,t}), \quad \forall g, \forall t \end{aligned} \quad (6)$$

The generation limits bound dispatched power for each committed unit:

$$P_g^{\min} X_{g,t} \leq P_{g,t} \leq P_g^{\max} X_{g,t}, \quad \forall g, \forall t \quad (7)$$

The DC power flow equations use the line susceptance $B_l = 1/x_l$:

$$\begin{aligned} F_{l,t} &= B_l(\theta_{i,t} - \theta_{j,t})P_N, \quad \forall l = (i,j), \forall t \\ |F_{l,t}| &\leq F_l^{\max}, \end{aligned} \tag{8}$$

The transition indicators satisfy:

$$u_{g,t} - v_{g,t} = X_{g,t} - X_{g,t-1}, \quad \forall g, \forall t \tag{9}$$

$$u_{g,t} + v_{g,t} \leq 1, \quad \forall g, \forall t \tag{10}$$

The startup cost uses the convex hull constraints $u_{g,t} \geq X_{g,t} - X_{g,t-1}$ and $u_{g,t} \geq 0$. These constraints give the cost term $C_{\text{st},g}u_{g,t}$ in the objective for each feasible binary commitment trajectory.